

An MIQCP-based multi-objective optimal operation strategy for renewables-integrated smart distribution systems considering transformer loss of life and environmental emissions

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ABSTRACT

Owing to growing electricity demand and environmental concerns, renewable-energy-based distributed generations (DGs) are widely integrated into today's distribution systems. However, high penetration of renewables DGs (RDGs) requires additional investments in high-cost step-up transformers connecting RDGs to the grid. On the other side, regarding variability of RDGs' output power, the connecting transformer's capacity is not completely utilized in a considerable time period. In this paper, a smart-grid technology-based scheduling model is proposed to efficiently utilize the free capacity of existing transformers without investing on new ones. The model is based on dynamic thermal rating (DTR) and loss of life (LOL) feature of transformer. The developed approach is formulated as a mixed-integer quadratic constrained programming (MIQCP) as a convex model to obtain global optimum solutions. The conducted model is treated as a multi-objective optimization with LOL minimization as well as emission reduction. The proposed multi-objective model is implemented using ϵ -constraint method where the final solution is selected by fuzzy-satisfying and max-min techniques. The model is implemented in GAMS and the optimization is accomplished by GUROBI solver to show efficacy of the proposed approach. The simulations demonstrate the capability of conducted approach in managing the CO₂ emission as well as transformer loss of life. Among different Pareto solutions, the best solution reduces the CO₂ emission by 11.7% while the transformer's aging is decreased by 19.2% demonstrating environmental, economic, and technical advantages of the proposed model.

1. Introduction

1.1. Background and motivation

Distribution networks have the task of receiving electricity from transmission systems at high voltage (HV) levels and delivering it to consumers at medium or low voltage (MV-LV) levels [1–3]. Due to radial topology of distribution systems, the network operators are usually engaged with high voltage drop and power losses. These problems are more highlighted when the load demands of consumers are grown [4]. To resolve the mentioned problems, different short-term and long-term solutions are employed by distribution network operators and planners, such as distribution network reconfiguration (DNR) [5,6], optimal capacitor banks placement [7,8], and reinforcement/installation of lines

(feeders) [9,10]. Another alternative is the use of DG units which are widely integrated in today's electric power grids [11]. DG units bring several advantages for distribution systems, such as reducing power losses and voltage drop, reliability enhancement, releasing capacity of equipment, and postponing the need for installation/reinforcement of lines and HV/MV transformers [12]. Among the DG units, the renewable-energy-based ones, like wind and photovoltaic (PV) units are under more attention due to their cleanliness and environmental-friendly features. They also relieve the concerns of countries regarding depletion of fossil-based fuels [13]. However, addition of more PV and wind capacities is subject to some limitations. An important limitation is the capacity of LV/MV step-up transformers connecting the wind and PV farms to the MV electric grid. Traditionally, expanding the capacity of PV and wind farms must be implemented in line with replacement of existing transformers with the ones having

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Nomenclature		
Indices		
i, j	Buses	
ij	Branches	
T	Time (hour)	
K	Linearization pieces	
Sets		
Ω_B	Set of buses	
Ω_{DTR}	Set of branches having DTR-equipped transformer	
Ω_t	Set of operating periods (hours)	
Constants		
Δt	Time resolution (hour)	
$V_{cut-in}, V_{cut-out}$	Cut-in and cut-out speeds of wind turbine (m/s)	
V_{rated}, V_t^w	Rated speed of wind turbine and wind speed at hour t (m/s)	
P_{rated}	Rated power of wind turbine (MW)	
$P_t^{WT,max}, P_t^{PV,max}$	Maximum available power of wind turbine/PV unit at hour t (MW)	
N_{PV}	Number of PV panels in photovoltaic farm (-)	
$V_{g,t}, I_{g,t}$	Output voltage and current of PV panel (V and A)	
FF	Filling factor of PV panel (-)	
V_{OC}	Open-circuit voltage of PV unit (V)	
I_{SC}	PV panel's short-circuit current (A)	
K_V, K_I	Temperature coefficient of voltage and current in PV panel (V/°C and A.m ² /W°C)	
$T_{cg,t}$	PV panel's temperature (°C)	
N_{OT}	PV panel's nominal operating temperature (°C)	
V_{MPP}, I_{MPP}	PV panel's maximum power point tracking voltage/current (V and A)	
SI_t	Solar radiation intensity at hour t (W/ m ²)	
T_t^{amb}	Ambient temperature at hour t (°C)	
$P_{i,t}^D, Q_{i,t}^D$	Active/reactive power of load demand at bus i and hour t (MW)	
em	CO ₂ emission of upward grid (kg/MWhr)	
$A_{\ell i}$	ℓi – th element of bus-line matrix for network. It is equal to “1” if bus i is the sending bus of line ℓ and “-1” if it is the receiving bus. Otherwise it is zero.	
$B_{\ell i}$	The same as $A_{\ell i}$ if “1” is replaced with “0”	
$R_{\ell}^{line}, X_{\ell}^{line}$	Resistance and reactance of line ℓ (Ω)	
U^{min}, U^{max}	Minimum and maximum value of square of voltage (per unit)	
τ_o, τ_w	Time constants of oil and winding in transformer (minutes)	
R	Ratio of load loss at rated current to the no-load loss of transformer (-)	
$\Delta T_{rated,ij}^{TO}, \Delta T_{ij,rated}^{HS}$	Top-oil and hot-spot rise at the rated load (°C)	
$I_{ij,rated}$	Rated current of branch ij (A)	
$\Delta J_{ij,t}^k$	Linearization pieces of square of current (A)	
$J_{ij}^{min}, J_{ij}^{max}$	Minimum and maximum values of J for linearization (A ²)	
n_1, n_2	Number of linearization pieces for top-oil and hot-spot temperatures (-)	
$S_{TO,ij}^k$	Slope of piece k in linearization of top-oil temperature (°C/A ²)	
$J_{ij,t,ini}^k, J_{ij,t,fin}^k$	Initial and final bounds for piece k of J in linearization (A ²)	
$\Delta T_{ij,t,k,fin}^{TO,f}$	Initial bound for piece k of $\Delta T_{ij,t}^{TO,f}$ in linearization (°C)	
$\Delta T_{ij,t,k,ini}^{TO,f}$	Final bound for piece k of $\Delta T_{ij,t}^{TO,f}$ in linearization (°C)	
$S_{HS,ij}^k$	Slope of piece k in linearization of hot-spot temperature (°C/A ²)	
$\Delta T_{ij,t,k,fin}^{HS,f}$	Initial bound for piece k of $\Delta T_{ij,t}^{HS,f}$ in linearization (°C)	
$\Delta T_{ij,t,k,ini}^{HS,f}$	Final bound for piece k of $\Delta T_{ij,t}^{HS,f}$ in linearization (°C)	
$S_{AF,ij}^k$	Slope of piece k in linearization of transformer's aging factor (hours/°C)	
$A_{ij,t,ini}^k, A_{ij,t,fin}^k$	Initial and final bounds for piece k of $A_{ij,t}$ in linearization (hours)	
$dT_{ij,t}^{HS,k}$	Linearization pieces of hot-spot temperature (°C)	
$T_{ij,t,fin}^{HS,k}, T_{ij,t,ini}^{HS,k}$	Initial and final bounds for piece k of hot-spot temperature in linearization (°C)	
Variables		
$P_{i,t}^{WT}, P_{i,t}^{PV}$	Wind and PV powers scheduled at bus i and hour t (MW)	
$AF_{ij,t}$	Aging factor for transformer connected between buses i and j at hour t (hours)	
$P_{i,t}^{grid}, Q_{i,t}^{grid}$	Active/reactive power received from the upward grid at bus i and hour t (MW)	
$P_{ij,t}^{net}, Q_{ij,t}^{net}$	Active and reactive power flows through branch ij without consideration of branch losses (MW)	
C_t	Ratio of current passing thorough the transformer at hour t to the rated current of the transformer (-)	
$\Delta T_t^{TO}, \Delta T_t^{HS}$	Top oil and hot-spot temperature rises at hour t (°C)	
$I_{ij,t}$	Current flowing in branch ij at hour t (A)	
$J_{ij,t}$	Square of current flowing in branch ij at hour t (A ²)	
$U_{i,t}$	Square of voltage of bus i at hour t (V ²)	
$\Delta T_{ij,t}^{TO,f}$	Final top-oil temperature rise at hour t for transformer of branch ij (°C)	
$\Delta T_{ij,t}^{HS,f}$	Final hot-spot temperature rise at hour t for transformer of branch ij (°C)	
$\alpha_{ij,t,k}$	Binary variable showing that if piece k is filled in linearization of top-oil and hot-spot temperatures	
$\beta_{ij,t,k}$	Binary variable showing that if piece k is filled in linearization of aging factor	
$T_{ij,t}^{HS}$	Hot-spot temperature of transformer ij at hour t (°C)	

higher nameplate capacity (rating), or with paralleling additional transformers [14]. On the other side, it should be noticed that as the output powers of wind and PV units are time-dependent regarding the variation of wind speed and solar radiation, their connecting transformer rarely reaches to its nameplate rating, and will have extra capacity during considerable time periods [15]. Therefore, addition of more transformer capacity with expansion of wind and PV farms capacity may not be economical. Instead, in today's power systems, introduction of smart-grid technologies (SGTs) has enabled network operators to have a real-time supervision and control on different equipment such as lines, switches, transformers, generating units, etc.

[16–20]. One of SGTs is dynamic transformer rating (DTR). The thermal rating of a transformer (in MVA) is dependent on its constructional characteristics (such as windings, iron core, cooling system, etc.) as well as weather parameters such as ambient temperature [21]. As the weather parameters are varied in different time, the rating will be also a variable quantity. However, the transformer designers report the thermal capacity on the “Nameplate” based on the worst-case weather condition (i.e. maximum ambient temperature). This rating is called as static transformer rating (STR) which is a conservative value that will occur only in short time periods [22]. If the transformer is equipped with SGTs, its real-time capacity, known as DTR, can be calculated in an Y

Table 1

Comparison of the proposed model with the previous researches.

Ref. No (Year)	Renewable Energy Sources		Network Constraints Consideration	Grid Level		LOL Consideration	Emission Consideration	Objective function		Optimization Model	
	Wind	PV		Transmission	Distribution			Single- objective	Multi- objective	Non- Convex	Convex
[25] (2019)	×	×	×	—	✓	✓	×	No optimization is applied		—	—
[26] (2020)	✓	×	DC-OPF	✓	×	✓	×	✓	×	Iterative Solving Process	
[22] (2020)	✓	×	DC-OPF	✓	×	×	×	✓	×	×	✓
[27] (2021)	×	×	×	—	✓	✓	×	✓	×	—	✓
[28] (2023)	✓	×	AC-OPF	✓	×	×	×	✓	×	—	✓
[29] (2022)	✓	×	AC-OPF	✓	×	×	×	✓	×	✓	×
[30] (2024)	✓	×	AC-OPF	✓	×	×	×	✓	×	—	✓
[31] (2020)	×	✓	AC-OPF	—	✓	✓	×	✓	×	—	✓
[14] (2023)	✓	×	×	✓	×	✓	×	No optimization is applied		—	—
[15] (2022)	✓	×	×	✓	×	×	×	✓	×	—	✓
[32] (2024)	✓	×	AC-OPF	✓	×	✓	×	✓	×	✓	×
Proposed model	✓	✓	✓	—	—	✓	✓	—	✓	—	✓

manner to be used in real-time scheduling of distribution system [23]. On the other side, if the transformers are not replaced by high-capacity ones when PV and wind farms are expanded, the loading of transformer will be increased even above its STR. Overloading of transformer brings about its temperature rise, insulation degradation, and possible loss of life [24]. Overloading and LOL of transformer are against expansion of renewable energy units and emission reduction. Therefore, it is an essential issue to smartly manage the transformer loading and temperature regarding penetration increase of renewable energy units. The current paper will address this issue and developed a model to simultaneously manage these two challenges. Subsequently, the related papers around this subject are reviewed to highlight the gap of existing researches and contributions of the present work.

1.2. Literature review

There have been different researches around dynamic thermal rating of transformer and its utilization in optimal operation of transmission and distribution systems. Ref. [25] presents a risk-based strategy for managing of distribution transformers based on their dynamic rating. The load and ambient temperature are provided as probabilistic inputs obtained from quantile regression forests. The insulation thermal ageing because of transformer overloading in terms of rewinding cost is compared with the cost of installing new transformer. This paper does not consider distribution system constraints and power flow equations, instead different scenarios of load profile are considered for the transformer.

To expand capacity of wind farms, Ref. [26] investigates the possibility of using DTR capability of step-up transformers without making an investment on new transformers. The transformer loss of life is converted to a cost to compare it with installation cost of new transformers. A single-node system along with a 6-node one followed by DC power flow equations are used to investigate the conducted approach. It is worth mentioning that the DC model of power flow does not reflect the network behavior appropriately as it disregards voltage and reactive power limitations.

Demand response (DR) and DTR are used in [27] to increase reserve of transformer. The reserve estimation based on a linear programming optimizes the DR volume and operation over a given time interval. The thermal limit, maximum power, and the aging effects of transformer are considered in the mathematical model. This paper ignores the electric grid topology and network-related constraints; i.e. the investigations are on a single-bus system.

Authors in [22] proposed a day-ahead scheduling model for power system integrated with large wind power plants. The model exploits DTR

as well as dynamic line rating (DLR) in order to minimize operation, load shedding, and wind power curtailment costs. The DC model of optimal power flow (DC-OPF) is utilized to consider system constraints. The results show positive effect of simultaneous usage of DLR and DTR. The work of [22] is extended in [28] to use AC optimal power flow (AC-OPF) for network calculations as well as to incorporate uncertainty of problem input parameters. Ref. [29] is the extended version of [22] which considers network-constrained units commitment instead of OPF considering AC power flow equations. The energy storage system (ESS) as well as phase shifting transformer (PST) are considered in [30] to develop the models of [22,28–30] and to exploit more capacity of wind farms and reduce the operational costs. However, none of the references [22, 28–30] have considered the transformer loss of life and insulation thermal aging.

Ref. [14] combines the thermal model of transmission lines and transformers to determine the potential increase in wind farm capacity. The impact of higher thermal limit on the LOL of the transformer is compared to the amount of curtailed wind and the potential revenue from the added wind farm. The network constraints are not considered in this paper.

Short-term and long-term planning of wind farm is implemented in [15] by the aid of DTR. Optimal wind farm capacity is determined by a mixed-integer linear programming (MILP) model. As the top-oil relation of transformer is nonlinear based on the IEEE Clause 7, its linearized model is employed to keep the model linear. The simulations obtain the relation between net present value of investment and the capacity of wind farm transformer. A sensitivity evaluation is accomplished to assess the consequences of wind speed, price of electricity, and lifelong of the transformer for the wind farm's feasibility. This paper does not consider LOL issues, and it disregards network-related constraints.

In [31], an optimal operation strategy is proposed for distribution networks incorporating DTR of transformer. Also, management of network flexibility options including on-load tap changer (OLTC), switchable capacitor banks, and static VAR compensator (SVCs) are coordinately implemented in order to minimize the total operation cost and to maximize the dispatched power of PV units connected to LV sides of the transformers. The model is a convex one with a single objective function composed of several cost items summed up with the weighting factors. The most recent research is [32], where dynamic line rating and dynamic transformer rating are co-optimized to obtain the most economic operational plan subject to lines' conductor as well as transformer's hot-spot temperature limitations. This work is in transmission network level where the optimization model is a single-objective non-linear programming (NLP).

1.3. Research gap and paper's contribution

Based on the literature, Table 1 compares in summary the features of the literature with the present work. In this paper, a convex MIQCP-based multi-objective network-constrained optimal operation strategy is proposed to simultaneously maximize scheduling of wind and PV units and minimize transformer loss of life. At first, the fundamental equations of dynamic transformer rating and its insulation aging are described and incorporated in AC-OPF model of distribution network. Then, the problem is formulated as a multi-objective optimization considering all the network, transformer, and renewable energies limitations. Afterwards, the non-linear terms are linearized to form a convex optimization model obtaining global optimum solution within GAMS environment. In brief, the contributions of this paper can be outlined as follows:

- Integration of DTR relations into optimal operation of smart distribution systems;
- Simultaneously considering transformer loss of life and environmental emissions using a multi-objective optimization;
- Linearizing non-linear relations to obtain a convex MIQCP optimization model;
- Considering all the distribution network constraints in the proposed operation model;
- Maximizing scheduling of wind and PV units as renewable energy sources along with life management of transformers.

2. Problem formulation

2.1. Dynamic thermal rating of transformer and loss of life characteristic

The loading of a transformer is constrained by respect to its temperature limitation and the cooling system. The temperature limitation is based on the hot-spot temperature (HST) which affects the insulation degradation and brings about transformer's loss of life (LOL). The IEEE loading guide C57.91 says that the HST, shown by T^{HS} , is expressed by (1) representing that the HST at each time step is depends on the ambient temperature (T_t^{amb}), top-oil temperature rise (ΔT_t^{TO}), and hot-spot temperature rise (ΔT_t^{HS}) at that time step [21,31]. ΔT_t^{TO} and ΔT_t^{HS} are denoted by relations (2) and (3) based on which the top-oil and hot-spot temperature rises are dependent on their values at the previous time steps by an exponential function. τ_o and τ_w are time constants of oil and winding, respectively, and Δt is the time step resolution, which is considered 1 hour in this paper. Final values of top-oil and hot-spot temperatures rise at hour t are obtained by (4) and (5). ΔT_{rated}^{TO} and ΔT_{rated}^{HS} are top-oil and hot-spot rises at the rated load. C_t indicates the ratio of current passing thorough the transformer at hour t to the rated current of transformer as (6). R shows the ratio of ohmic loss at rated current to the no-load loss of the transformer, and n and m are two constants depending on the transformer's cooling type. The aging factor (AF) for the transformer insulation is assessed by (7) as an exponential function of HST. If the HST is limited to its maximum value (110 °C), the AF will be lower than 1 hour. Otherwise, the aging factor becomes a value larger than 1 hour. For example for the HSTs of 100 °C and 120 °C, the aging factor becomes 0.35 hours and 2.71 hours, respectively. It is recommended that the sum of aging factors during one operating day does not exceed 24hours.

$$T_t^{HS} = T_t^{amb} + \Delta T_t^{TO} + \Delta T_t^{HS} \quad (1)$$

$$\Delta T_t^{TO} = (\Delta T_t^{TO} - \Delta T_{t-1}^{TO}) \times \left(1 - e^{-\frac{\Delta t}{\tau_o}}\right) + \Delta T_{t-1}^{TO} \quad (2)$$

$$\Delta T_t^{HS} = (\Delta T_t^{HS} - \Delta T_{t-1}^{HS}) \times \left(1 - e^{-\frac{\Delta t}{\tau_w}}\right) + \Delta T_{t-1}^{HS} \quad (3)$$

$$\Delta T_t^{TO} = \Delta T_{rated}^{TO} \times \left(\frac{1 + C_t^2 \times R}{1 + R}\right)^n \quad (4)$$

$$\Delta T_t^{HS} = \Delta T_{rated}^{HS} \times C_t^{2 \times m} \quad (5)$$

$$C_t = \frac{I_t}{I_{rated}} \quad (6)$$

$$AF_t = \exp\left(\frac{15000}{383} - \frac{15000}{T_t^{HS} + 273}\right) \quad (7)$$

2.2. Wind and PV power models

Regarding the power-speed curve of the wind turbine, the maximum extractable wind power is obtained as (8), where V_{cut-in} , $V_{cut-out}$ and V_{rated} are the cut-in, cut-out, and rated speeds of the wind turbine. Also, P_{rated} describes the rated power of the wind turbine, and V_t^w is the wind speed at hour t [33]. It should be that the output power of the wind turbine (between cut-in and rated speeds) can be more accurately modeled as a cubic function of the wind speed [34]. However, in this paper, like many other researches [23,29,33], it has been modeled as a liner function as an approximation. Off course, wind speed and power are the inputs of the proposed model, and the model can handle any desired input.

$$P_t^{WT,max} = \begin{cases} 0 & V_t^w < V_{cut-in} \text{ or } V_t^w \geq V_{cut-out} \\ P_{rated} \times \frac{V_t^w - V_{cut-in}}{V_{rated} - V_{cut-in}} & V_{cut-in} \leq V_t^w < V_{rated} \\ P_{rated} & V_{rated} \leq V_t^w < V_{cut-out} \end{cases} \quad (8)$$

For a PV farm, the maximum extractable power depends on its structural parameters as well as ambient temperature (T_t^{amb}) and solar radiation intensity (SI_t) as (9)-(13). For convenience, the parameters have been defined in the nomenclature [35].

$$P_t^{PV,max} = N_{PV} \times FF \times V_{g,t} \times I_{g,t} \quad (9)$$

$$FF = \frac{V_{MPP} \times I_{MPP}}{V_{OC} \times I_{sc}} \quad (10)$$

$$T_{cg,t} = T_t^{amb} + SI_t \times \left(\frac{N_{OT} - 20}{0.8}\right) \quad (11)$$

$$I_{g,t} = SI_t \times [I_{sc} + K_I(T_{cg,t} - 25)] \quad (12)$$

$$V_{g,t} = V_{oc} - K_V T_{cg,t} \quad (13)$$

2.4. Proposed scheduling model

The proposed problem is a multi-objective scheduling optimization of smart distributions system integrated with PV and wind farms by considering DTR and LOL of transformers connecting the farms to the network.

2.4.1. Objective functions

The first considered objective function includes minimization of environmental emission as shown in (14). It includes emission of CO2 corresponding to the power received by distribution system from the upward grid. The upward grid is the transmission system within which the centralized thermal generators are located. The thermal generating units produce greenhouse gases for the aim of electricity generation, and cause CO2 emission. P_{it}^D , P_{it}^{WT} , and P_{it}^{PV} are respectively the active power

of load demand, output power of wind unit, and output power of PV unit located at bus i and hour t . em is the emission value of CO₂ in kg/MWhr.

$$OF_1 = \text{Min} \left\{ \text{Emission} = em \times \left(\sum_{i \in \Omega_B} \sum_{t \in \Omega_t} P_{i,t}^D - (P_{i,t}^{WT} + P_{i,t}^{PV}) \right) \right\} \quad (14)$$

The second objective function is minimization of loss of life of the transformer as (15). The LOL is in fact, the sum of aging factors obtained by (7) for all the DTR-equipped transformers at all hours of the operating day. Ω_{DTR} is the set of DTR-equipped transformers, and ij is the branch in which the transformer is located (between buses i and j).

$$OF_2 = \text{Min} \left\{ \text{LOL} = \sum_{ij \in \Omega_{DTR}} \sum_{t=1}^T AF_{ij,t} \right\} \quad (15)$$

2.4.2. Problem constraints

• Power flow relations

The AC power flow equations are the most important limitations among distribution network constraints. These equations are essential to give the system operator information about operation condition including the loading of lines and transformers as well as voltage of buses. The traditional power flow equations are non-linear causing the optimization model to be non-convex. In this regard, the line flow based (LFB) model of load flow, as convex formulation, has been used in this paper according to (16)-(23) [9]. The parameters used in the relations have been defined in the nomenclature. In the LFB power flow, instead of voltages of buses ($V_{i,t}$) and currents of branches ($I_{ij,t}$), their square values have been used as $U_{i,t}$ and $J_{ij,t}$. In this model, based on the Kirchhoff's laws, the magnitudes of voltages and currents are employed and their angles are not appeared. $P_{ij,t}^{net}$ and $Q_{ij,t}^{net}$ are the active and reactive power flows without consideration of branch losses. $B_{\ell i} R_{\ell}^{line} J_{ij,t}$ and $B_{\ell i} X_{\ell}^{line} J_{ij,t}$ show the branches' active and reactive power losses, respectively.

$$P_{i,t}^{grid} + P_{i,t}^{WT} + P_{i,t}^{PV} - P_{i,t}^D = \sum_{j \in \Omega_B} A_{\ell i} P_{ij,t}^{net} + \sum_{j \in \Omega_B} B_{\ell i} R_{\ell}^{line} J_{ij,t}; \forall t \in \Omega_t, i, j \in \Omega_B, \ell = ij \quad (16)$$

$$Q_{i,t}^{grid} - Q_{i,t}^D = \sum_{j \in \Omega_B} A_{\ell i} Q_{ij,t}^{net} + \sum_{j \in \Omega_B} B_{\ell i} X_{\ell}^{line} J_{ij,t}; \forall t \in \Omega_t, i, j \in \Omega_B, \ell = ij \quad (17)$$

$$U_{i,t} - U_{j,t} + 2(R_{\ell}^{line} P_{ij,t}^{net} + X_{\ell}^{line} Q_{ij,t}^{net}) + J_{\ell,t} [(R_{\ell}^{line})^2 + (X_{\ell}^{line})^2] = 0; \forall t \in \Omega_t, i, j \in \Omega_B, \ell = ij \quad (18)$$

$$P_{ij,t}^{net} = -P_{ij,t}^{net}; \forall t \in \Omega_t, i, j \in \Omega_B \quad (19)$$

$$Q_{ij,t}^{net} = -Q_{ij,t}^{net}; \forall t \in \Omega_t, i, j \in \Omega_B \quad (20)$$

$$P_{ij,t}^{net^2} + Q_{ij,t}^{net^2} \leq J_{ij,t} U_{ij,t}; \forall t \in \Omega_t, i, j \in \Omega_B \quad (21)$$

$$J_{ij,t} = I_{ij,t}^2; \forall t \in \Omega_t, i, j \in \Omega_B \quad (22)$$

$$U_{i,t} = V_{i,t}^2; \forall t \in \Omega_t, i \in \Omega_B \quad (23)$$

• Buses voltage limit and branches current limit

Regarding the LFB model of AC power flow, the voltage magnitude of buses must satisfy relation (24). Also, the current passing through the branches must be limited as (25).

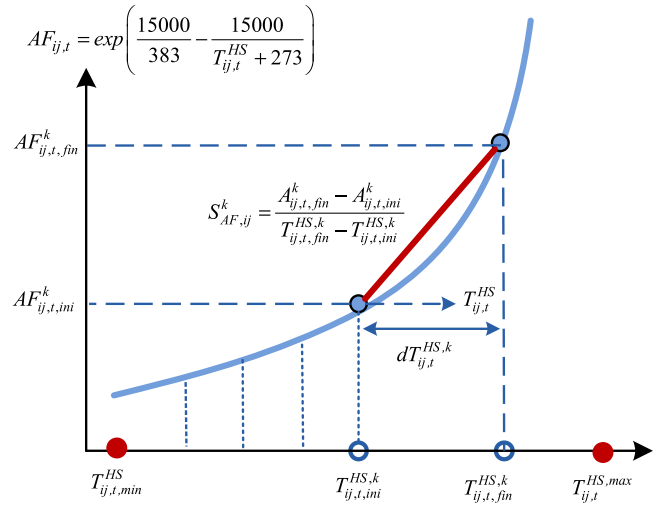


Fig. 1. Piecewise linearization of aging factor of transformer in terms of hot-spot temperature.

$$U^{\min} \leq U_{i,t} \leq U^{\max}; \forall t \in \Omega_t, i \in \Omega_B \quad (24)$$

$$J_{ij,t} \leq J_{ij}^{\max}; \forall t \in \Omega_t, i, j \in \Omega_B \quad (25)$$

• Wind and PV units constraints

The wind and PV powers' equations were described in (8) and (9). In the proposed OPF model, the scheduled powers of wind and PV farms must follow the limitations of (26) and (27), where $P_{i,t}^{WT,max}$ and $P_{i,t}^{PV,max}$ are the available wind and PV powers at bus i and hour t .

$$P_{i,t}^{WT} \leq P_{i,t}^{WT,max}; \forall t \in \Omega_t, i \in \Omega_B \quad (26)$$

$$P_{i,t}^{PV} \leq P_{i,t}^{PV,max}; \forall t \in \Omega_t, i \in \Omega_B \quad (27)$$

• DTR-equipped Transformers

For the step-up transformers that connect the wind and PV farms to the distribution grid, the DTR capability is considered. For these transformers, the relations (1)-(7) must be satisfied. However as the inclusion of these non-linear relations into the proposed OPF problem makes the model non-convex, subsequently, these equations will be linearized to make a convex model.

2.4.3. Model convexification

Among the relations describing DTR model of transformer and its LOL, the relations (4), (5), and (7) are non-linear. To make the model convex, the piece-wise linearization (PWL) [28,36] is employed. From (6) and (22), the relation (4) and (5) can be reformulated as (28) and (29) for the transformer connected between buses i and j in terms of square of current, $J_{ij,t}$.

$$\Delta T_{ij,t}^{TO,f} = \Delta T_{rated,ij}^{TO} \times \left(\frac{1 + \frac{J_{ij,t}}{I_{ij,rated}^2} \times R}{1 + R} \right)^n \quad (28)$$

$$\Delta T_{ij,t}^{HS,f} = \Delta T_{rated,ij}^{HS} \times \left(\frac{J_{ij,t}}{I_{ij,rated}^2} \right)^m \quad (29)$$

By using the PWL, the relation (28) is expressed as (30)-(34). Also Eq. (29) is linearized using (30), (31), (35), and (36). In the similar way, the

relation of (7) for the aging factor of the transformer is linearized using (37)-(40). The concept of piecewise linearization has been typically shown in Fig. 1 for the aging factor. In the relations, $\alpha_{ij,t,k}$ and $\beta_{ij,t,k}$ are binary variables indicating that if piece k is filled in the linearization process or not.

After the mentioned linearizations and considering the objective function and LFB power flow, the proposed model becomes a mixed-integer quadratic constrained programming (MIQCP) as a convex optimization. As mentioned before, solving the convex models obtains global optimum solution for the problem.

$$\Delta J_{ij,t}^k = \frac{J_{ij,t}^{\max} - J_{ij,t}^{\min}}{n_1}; \forall ij \in \Omega_{DTR}, \forall t \in \Omega_t \quad (30)$$

$$J_{ij,t,fin}^k = J_{ij,t,min}^k + \Delta J_{ij,t}^k; \forall ij \in \Omega_{DTR}, \forall t \in \Omega_t, \forall k \in 1, 2, \dots, n_1 \quad (31)$$

$$S_{TO,ij}^k = \frac{\Delta T_{ij,t,k,fin}^{TO,f} - \Delta T_{ij,t,k,ini}^{TO,f}}{J_{ij,t,fin}^k - J_{ij,t,ini}^k}; \forall ij \in \Omega_{DTR}, \forall t \in \Omega_t, \forall k \in 1, 2, \dots, n_1 \quad (32)$$

$$J_{ij,t} = J_{ij,t,min} + \sum_{k=1}^{n_1} \alpha_{ij,t,k} \times \Delta J_{ij,t}^k; \forall ij \in \Omega_{DTR}, \forall t \in \Omega_t \quad (33)$$

$$\Delta T_{ij,t}^{TO,f} = \Delta T_{ij,t,min}^{TO,f} + \sum_{k=1}^{n_1} \alpha_{ij,t,k} \times S_{TO}^k \times \Delta J_{ij,t}^k; \forall ij \in \Omega_{DTR} \quad (34)$$

$$S_{HS,ij}^k = \frac{\Delta T_{ij,t,k,fin}^{HS,f} - \Delta T_{ij,t,k,ini}^{HS,f}}{J_{ij,t,fin}^k - J_{ij,t,ini}^k}; \forall ij \in \Omega_{DTR}, \forall t \in \Omega_t, \forall k \in 1, 2, \dots, n_1 \quad (35)$$

$$\Delta T_{ij,t}^{HS,f} = \Delta T_{ij,t,min}^{HS,f} + \sum_{k=1}^{n_1} \alpha_{ij,t,k} \times S_{HS}^k \times \Delta J_{ij,t}^k; \forall ij \in \Omega_{DTR}, \forall t \in \Omega_t \quad (36)$$

$$dT_{ij,t}^{HS,k} = \frac{T_{ij,t}^{HS,max} - T_{ij,t}^{HS,min}}{n_2}; \forall ij \in \Omega_{DTR}, \forall t \in \Omega_t \quad (37)$$

$$S_{AF,ij}^k = \frac{A_{ij,t,fin}^k - A_{ij,t,ini}^k}{T_{ij,t,fin}^{HS,k} - T_{ij,t,ini}^{HS,k}}; \forall ij \in \Omega_{DTR}, \forall t \in \Omega_t, \forall k \in 1, 2, \dots, n_2 \quad (38)$$

$$T_{ij,t}^{HS} = T_{ij,t,min}^{HS} + \sum_{k=1}^{n_2} \beta_{ij,t,k} \times dT_{ij,t}^{HS,k}; \forall ij \in \Omega_{DTR}, \forall t \in \Omega_t \quad (39)$$

$$AF_{ij,t} = AF_{ij,t,min} + \sum_{k=1}^{n_2} \beta_{ij,t,k} \times S_{AF,ij}^k \times dT_{ij,t}^{HS,k}; \forall ij \in \Omega_{DTR}, \forall t \in \Omega_t \quad (40)$$

2.4.4. Multi-objective optimization

The proposed model in this paper is a multi-objective optimization including minimization of environmental emission and minimization of loss of life of the transformers. These two objectives are contradictory. There are two approaches for handling of multi-objective optimization. One is to convert it to a single-objective one using weighting factors [22, 28,30], and the other is to provide Pareto-optimal solutions [36,37]. For the latter case, normal boundary intersection (NBI) and ϵ -constraint techniques are two well-known methods to prepare Pareto-optimal solutions [38]. These techniques are generally similar to each other in transforming the multi-objective problems to a series of single optimization problems. Their difference is in distribution of Pareto-optimal solutions within the search space. In this paper, ϵ -constraint method is employed to obtain Pareto-optimal solutions within GAMS environment. In this method, one of the objective functions is selected to be optimized while the other(s) are converted into additional constraints, leading to a solution that can be proven to always be Pareto-optimal [38]. For the aim of multi-objective optimization in this paper, an iterative process is implemented as the following steps to minimize the two objective functions given in (14) and (15) as OF_1 and OF_2 .

Step 1) Minimize OF_1 given in (14) considering all the problem constraints, and:

1-1) Set $of_1^{\min}$ as the optimum value of OF_1

1-2) Set $of_2^{\max}$ as the value of OF_2 when OF_1 is optimized

Step 2) Minimize OF_2 given in (15) considering all the problem constraints, and:

2-1) Set $of_2^{\min}$ as the optimum value of OF_2

2-2) Set $of_1^{\max}$ as the value of OF_1 when OF_2 is optimized

Step 3) Perform the following loop:

3-1) Set counter $c=1$

3-2) calculate epsilon as $\epsilon = of_2^{\min} + \frac{of_2^{\max} - of_2^{\min}}{c_{\max} - 1} \times (c - 1)$

3-3) Minimize OF_1 considering all the problem constraints as well as the following constraint:

$$OF_2 \leq \epsilon$$

3-4) Save the Pareto-optimal solution number c .

3-5) $c = c + 1$

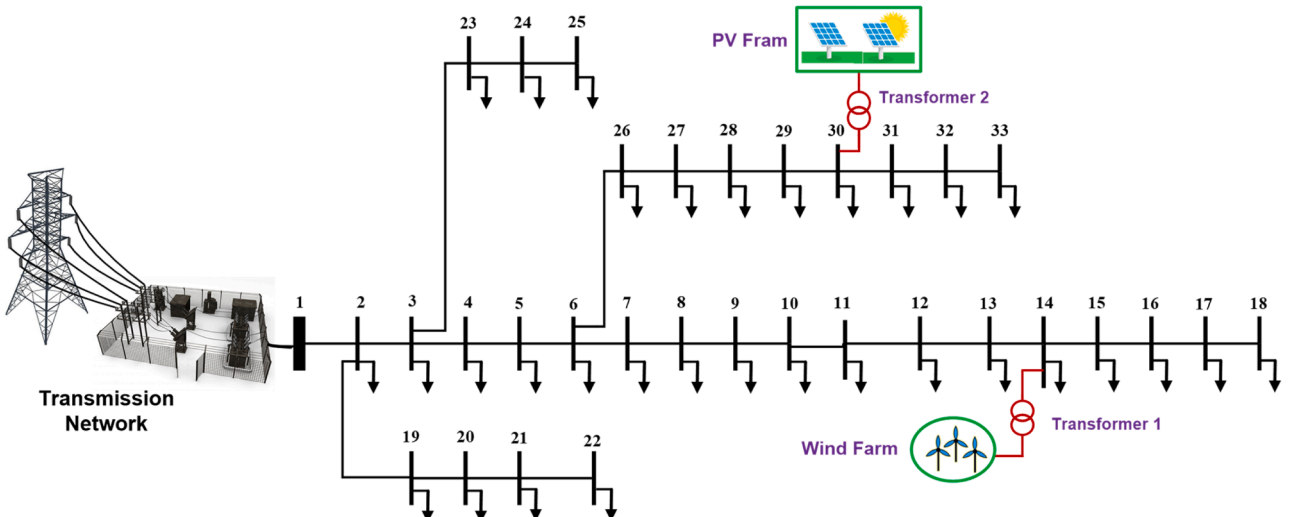


Fig. 2. Single-line diagram of the test system.

Table 2
Characteristics of PV farm.

Parameter	Value	Parameter	Value
V_{MPP}	28.36 (V)	N_{OT}	43 (°C)
I_{MPP}	7.76 (A)	N_{PV}	10,000
V_{OC}	36.96 (V)	K_V	0.1278 (V/°C)
I_{sc}	8.38 (A)	K_I	0.00545(A/°C)

3–6) If $c < c^{\max}$ go to step 3–2

Step 4) List all the values of OF_1 and OF_2 for all the values of c from 1 to $c^{\max}$.

Each of the obtained Pareto solutions is optimal its own, and the planner can choose any of them. However, if it is aimed to choose a solution being appropriate from the viewpoints of all objective functions, the fuzzy satisfaction method and max-min techniques are utilized to choose the final solution. The fuzzy satisfaction method is very effective due to its simplicity and similarity to human judgment in decision making. In this method, fuzzy satisfaction values are attributed to each of the solutions (each of the Pareto-optimal solutions) by the membership function. The membership function is a uniformly decreasing one with upper and lower bounds according to (41). In (41), x_c is the index of each solution, and $f_k(x_c)$ is the k -th objective function (criterion) of solution x_c . $\mu_k^{f_k(x_c)}$ is the membership function of each solution related to k -th objective function, and $f_k^{\min}$ and $f_k^{\max}$ are the minimum and maximum values of k -th criterion. The closer the objective function is to the minimum value, the closer its fuzzy value is to 1 and vice versa. After determining the membership function for each solution, the planner must determine the degree of satisfaction of each criterion and then select the final solution with the help of an optimization method. In this study, the max-min optimization method has been used to select the final solution, where its relationship is in the form (42) [36]:

$$\mu_k^{f_k(x_c)} = \begin{cases} 0 & \text{otherwise} \\ \frac{f_k^{\max} - f_k(x_c)}{f_k^{\max} - f_k^{\min}} & f_k^{\min} \leq f_k(x_c) \leq f_k^{\max} \end{cases} \quad (41)$$

$$\mu^{\max} = \max_{k=1:2} (\min \mu_k^{f_k(x_c)}) \quad (42)$$

2.4.5. Optimization tool

GAMS is a powerful software package for optimizing various problems with variety of variables and constraints. Having suitable solvers, this software is able to find the solutions of linear and non-linear optimization models with real and binary variables [33]. The basic model of the proposed problem was non-linear and non-convex, which is hard to solve, and its solution does not guarantee the global optimum point. By the aid of the proposed convexifications, the model has been converted to a mixed-integer quadratic constrained programming (MIQCP), which is convex model. In this paper, the “GUROBI” solver in the GAMS is employed to find the global optimum solution for the proposed model [39].

3. Numerical study

3.1. Describing the test network

The analyses of this paper are performed on the IEEE 33-bus system

Table 3
Thermal parameters of transformers.

Nameplate Rating		τ_o	τ_w	$\Delta\theta^{\text{top, rated}}$	$\Delta\theta^{\text{hst, rated}}$	R
Transformer 1	Transformer 2					
500kVA	750kVA	155minutes	5minutes	55 °C	25 °C	8

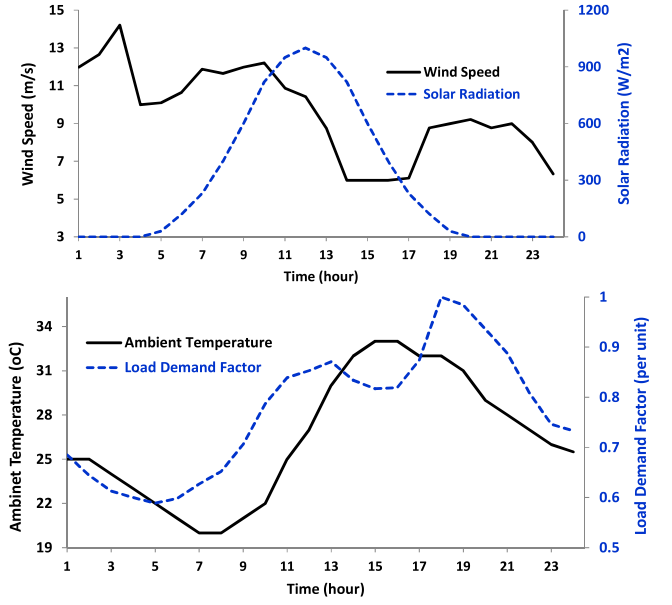


Fig. 3. Hourly profile of load demand and weather parameters.

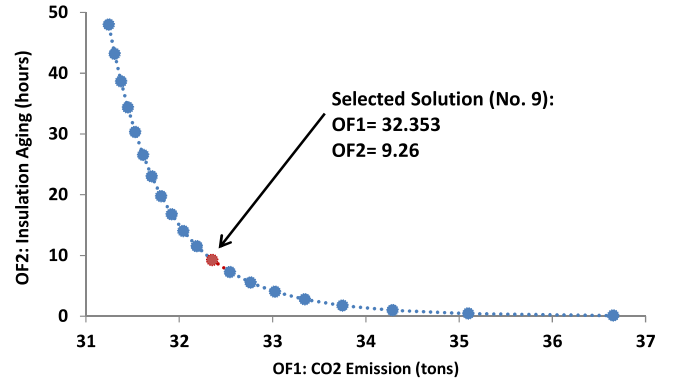


Fig. 4. Pareto-optimal solutions and the selected solution using fuzzy satisfaction technique.

depicted in Fig. 2. The system is a 12.66kV network at the MV level and 400V at the low-voltage level. The peak load demand of network is 3.715MW and 2.3MVar. The complete data related to the branches and loads of this network is given in [40]. There are two renewable energy sources in this network at low-voltage (LV) level, which is 400V. A PV farm with the characteristics of Table 2 is connected to bus 30. Also, a wind farm with the rated power of 1MW is attached to bus 14. The rated, cut-in, and cut-out speeds of the wind farm are 14m/s, 4m/s, and 25m/s, respectively. The wind and PV farms are connected to the medium voltage (MV) side through two 400V/12.66kV DTR-equipped step-up transformers, named as Transformers 1 and 2. Table 3 shows the parameters for the thermal model of the two transformers. As the original network is exposed to high voltage drop, two 600kVar capacitor banks are installed at buses 14 and 30 to compensate for the reactive power. The hourly profile of the wind speed, solar radiation, ambient temperature, and load demand are given in Fig. 3.

Table 4
Details of Pareto-optimal solutions obtained by ε -constraint method.

Solution No.	OF_1 (tons)	OF_2 value (hours)	μ^{OF_1}	μ^{OF_2}	$\mu^{\min}$
1	36.653	0.1	0	1	0
2	35.099	0.417	0.287	0.993	0.287
3	34.289	0.962	0.437	0.982	0.437
4	33.751	1.741	0.537	0.966	0.537
5	33.348	2.757	0.611	0.945	0.611
6	33.028	4.015	0.67	0.918	0.67
7	32.766	5.516	0.719	0.887	0.719
8	32.544	7.264	0.76	0.85	0.76
9*	32.353	9.26	0.795	0.809	0.795
10	32.188	11.505	0.826	0.762	0.762
11	32.045	14.003	0.852	0.71	0.71
12	31.919	16.753	0.875	0.652	0.652
13	31.806	19.757	0.896	0.59	0.59
14	31.705	23.017	0.915	0.522	0.522
15	31.613	26.534	0.932	0.448	0.448
16	31.528	30.309	0.948	0.369	0.369
17	31.449	34.343	0.962	0.285	0.285
18	31.376	38.636	0.976	0.195	0.195
19	31.307	43.19	0.989	0.1	0.1
20	31.245	48	1	0	0

3.2. Simulation results

The proposed model is applied to the test system in the form of multi-objective optimization using ε -constraint technique. The number of Pareto-optimal solutions has been considered to be 20 ones. Fig. 4 depicts the mentioned solutions in terms of the two objective functions. Also, the fuzzy satisfaction values of the solutions are provided in Table 4. It is seen that the initial solutions have lower values of aging

factor and higher values of CO₂ emission, while the ending solutions have high aging factor and low CO₂ emission. Based on the max-min rule, the solution No. 9 is chosen as the final solution. There is a reasonable trade-off among the OF_1 and OF_2 for this solution. The CO₂ emission for this solution is 32.353 tons, and the aging factor is 9.26 hours. It should be noted that the value of OF_2 is the sum of insulation aging for the two transformers shown in Fig. 2. By comparing solution No. 9 with solution No. 1, it can be seen that the CO₂ emission (OF_1) is reduced from 36.653 tons to 32.353 tons by 11.73%. From solution No. 9 to No. 20, the emission reduction is 3.42%; this is while the aging (OF_2) is increased from 9.26 hours to 48 hours by 418.36%. That is, for reducing a little value of CO₂ emission, the aging of transformers becomes 5.1 times. Therefore, the solution No. 6 provides a reasonable trade-off between emission reduction and insulation aging reduction; that is, for solution No. 9, almost the same scheduling of renewable energy sources (the same CO₂ emission) as solution No. 20 is fulfilled by about 19.2% of insulation aging.

The hourly values of scheduled wind and PV powers have been compared in Figs. 5 and 6 for the solutions 1, 9, and 20. It can be observed that the values of wind and PV power curtailment for solution No. 1 are high; in the other words, the scheduled wind and powers are low. For the solution No. 20, the renewables scheduling is increased considerably, but the aging of transformers is accordingly high, that is not favorable. For the selected solution No. 9, the wind and PV powers scheduling is near to solution No. 20, however, the aging is very low. As mentioned, the solution No. 9 provides a promise between maximization of renewables scheduling (minimization of CO₂ emission) and maximization of the life of transformer (minimization of insulation aging).

The hot-spot temperature (HST) of transformer-1 (connected

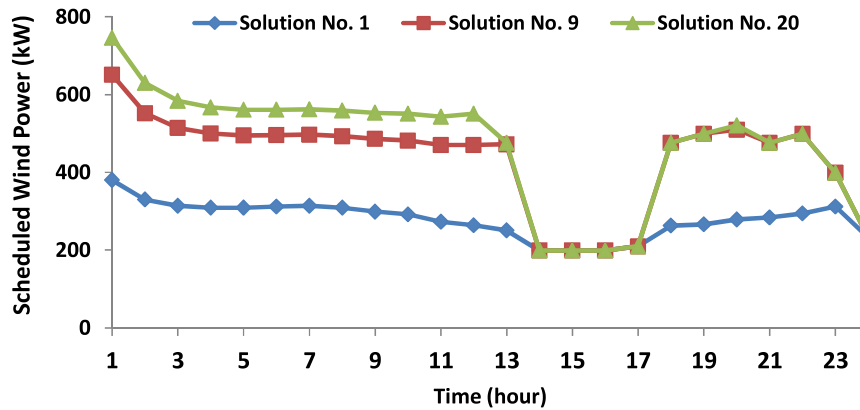


Fig. 5. Comparing the scheduled wind power for three solutions.

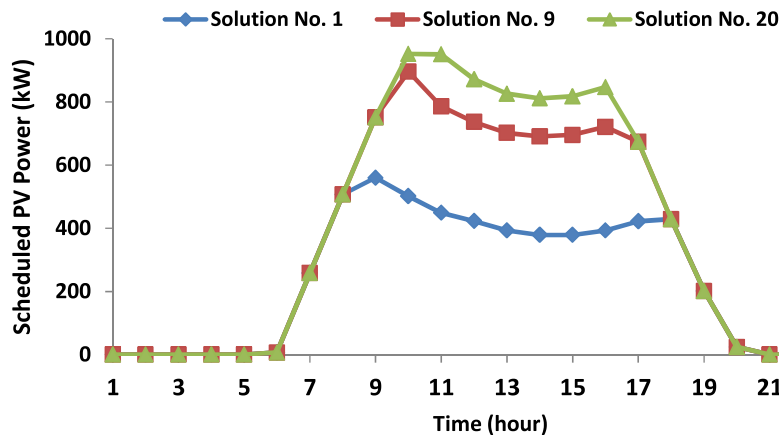


Fig. 6. Comparing the scheduled PV power for three solutions.

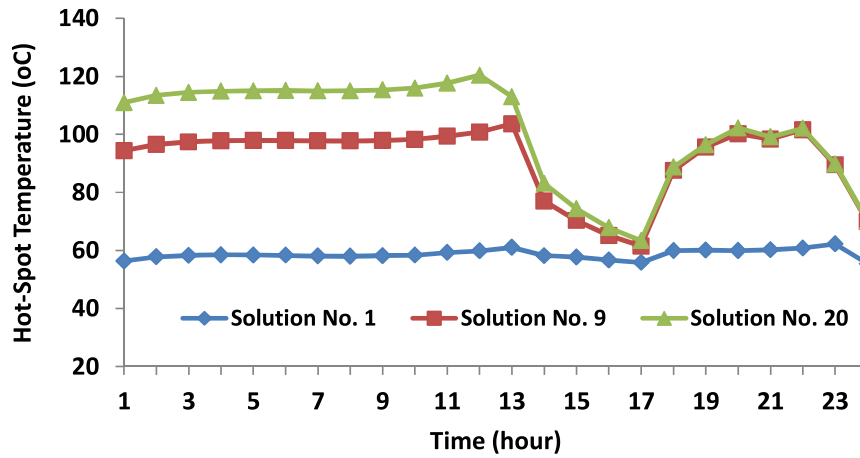


Fig. 7. Comparing the hot-spot temperature of Transformer-1 in three solutions.

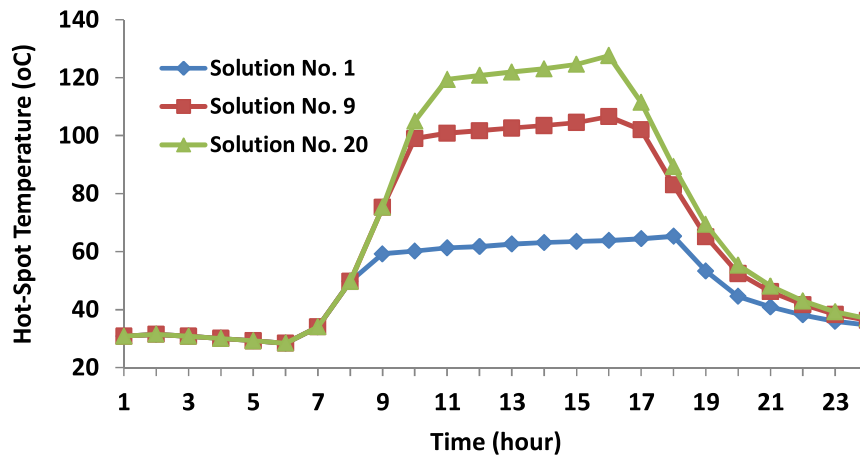


Fig. 8. Comparing the hot-spot temperature of Transformer-2 in three solutions.

between buses 14 and 34) and transformer -2 (connected between buses 30 and 35) have been depicted in Figs. 7 and 8. In Fig. 7, for the solution No. 1, the HST is at most 62°C. For solution No. 20, the HST exceeds the temperature of 120°C at some hours. However, for the selected solution No. 9, the maximum value of HST is about 103°C which occurs at hour 13. In other words, for the selected solution, the HST does not reach 110°C (as the temperature usually considered by power system operators). A similar discussion can be made about Fig. 8 for the transformer 30–35.

The aging factors and their accumulated values have been depicted in Fig. 9 for the transformer-1. It is seen that the higher aging factors occur mostly at the hours that the HST is high. The loading of the transformer-1 is also given in Fig. 10 indicating some hours with higher loadings. For the solutions No. 9 and No. 20, the overloading of transformers (loadings larger than 1 per-unit) can be observed at some hours; however, this overloading does not impact the overall aging (sum of the aging factors in one day) of transformer more than usual (24 hours). In other words, high loadings and aging factors are compensated at the next hours to prevent reduction of transformer's life more than the permitted value.

It should be noted that all the advantages obtained from the dynamic transformer rating are along with satisfaction of all the network operating constraints. For example, for the selected solution No. 9, the voltage magnitudes of all buses at all operating hours are graphically shown in Fig. 11. As seen, at all 24 hours of the day, the permitted ranges of voltage, i.e., [0.95pu-1.05pu] has been satisfied. The minimum voltage level is around the hours 19 to 21. In this time interval, the

maximum load demand is occurred, the PV units do not inject active power, and the wind speed (wind power) is lower than other hours.

5. Conclusion

This paper proposed a smart-grid technology-based optimal operation strategy to coordinately manage renewable energy sources along with transformers loss of life. The developed model is formulated as a multi-objective optimization to provide Pareto-optimal solutions using ϵ -constraint method. For selection of the final solution, fuzzy-satisfaction and max-min techniques are employed to make a promise between CO₂ emission reduction and reduction of transformers' insulation aging. The basic model was non-linear and non-convex, but using the convexifications and linearizations, the optimization problem was transformed to a convex MIQCP model whose solution in GAMS provides global optimum solutions. Applying the developed model on the IEEE 33-bus network demonstrated that by smart scheduling, distribution system operator can appropriately schedule the wind and PV farms without decreasing the insulation life of the transformer. It was demonstrated that almost the same scheduling of PV and wind farms (the same CO₂ emission) is fulfilled with about 19.2% of insulation aging, while the CO₂ emission is decreased by 11.7%. In this way, a clean as well as economical operation is fulfilled for the distribution system. The uncertainty of environmental parameters including wind speed, solar radiation, and ambient temperature may affect the results obtained in this paper. These uncertainties can be modeled by different probability distribution functions (PDFs) such as Weibull, Beta, and

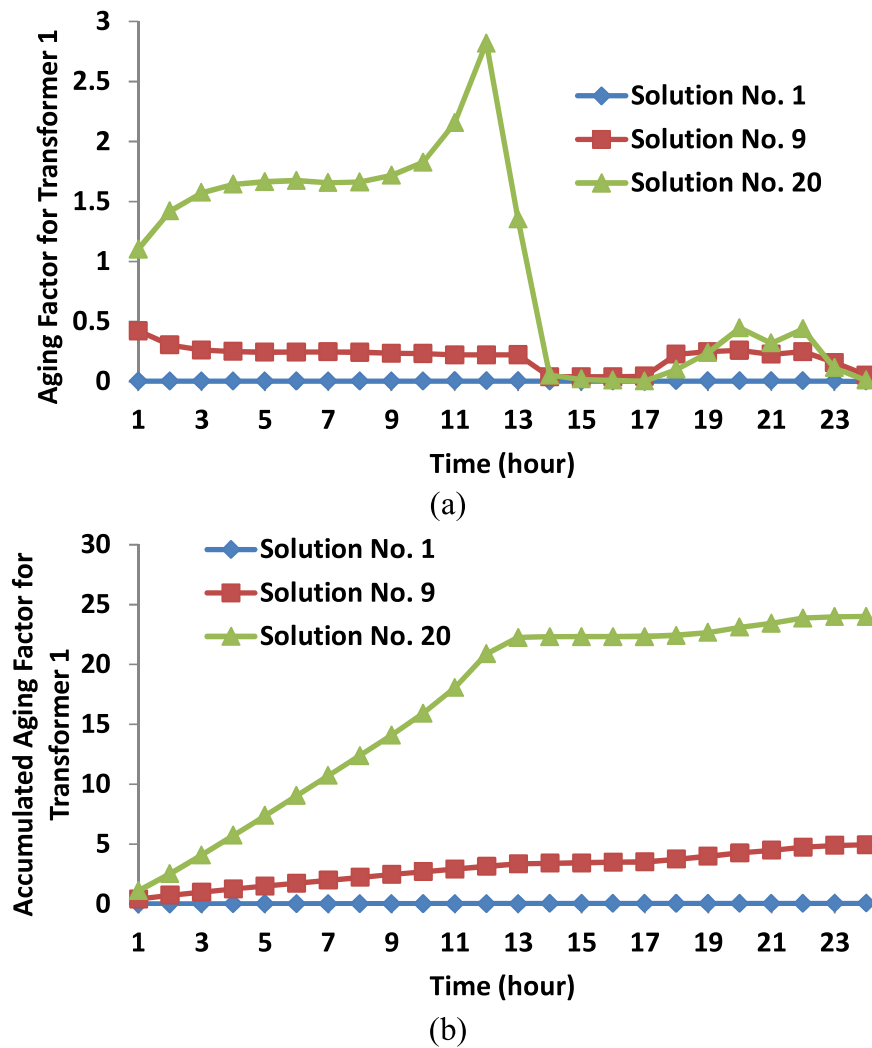


Fig. 9. Aging factor of Transformer-1: (a)-Individual values, (b)-Accumulated values.

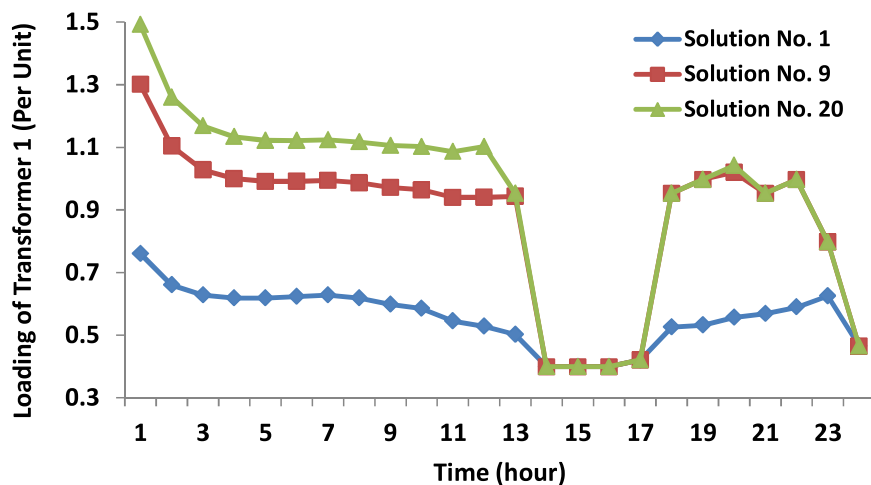


Fig. 10. Loading of transformers 1 in three solutions.

normal PDFs [28,39]. Also, the uncertainties can be handled by Monte-Carlo simulation, scenario-based programming, robust optimization, point estimate method (PEM), etc. As the proposed model in this paper focuses on the effect of DTR and loss of life on capturing more renewable energies and CO₂ emission reduction, the uncertainty has not been considered. This issue will be addressed in a future research.

CRediT authorship contribution statement

Amir Bagheri: Writing – original draft, Validation, Investigation, Data curation, Conceptualization. **Saleh Mobayen:** Writing – review & editing, Validation, Project administration, Formal analysis. **Nasrin Osali:** Visualization, Software, Resources, Methodology.

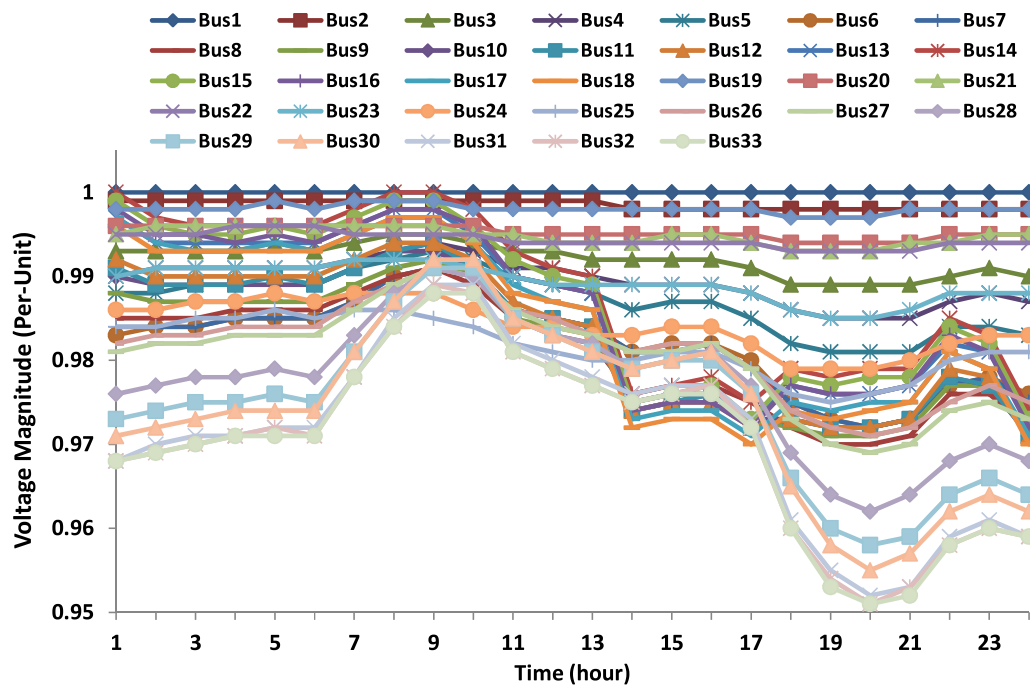


Fig. 11. Voltage magnitude of all buses at different hours of the day.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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